Bow Valley College

Course Outline

DATA3999

Data Management and Analytics Capstone Project
Winter 2023 - Spring 2025

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DATA3999 Data Management and Analytics Capstone Project

COURSE DESCRIPTION

Working alone or in a small team, students research, design, develop, and implement an applied big data analytics research project to satisfy a real organizational or community need. Students are expected to apply all of their knowledge and skills to produce a functioning prototype of their project idea.

REQUISITES	None
EQUIVALENTS	None
CREDITS	6
HOURS	90
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	

COURSE COMPETENCY

COMPETENCY TITLE

Plan, Execute and Present Data Analysis Projects

COMPETENCY STATEMENT

Data Management and Analytics professionals apply business analysis, data analysis, and design skills to plan, execute, and present data-driven projects that solve business problems or identify business opportunities.

COMPETENCY DESCRIPTION

With the emergence of data and data-driven solutions, data management and analytics professionals fill critical job roles across many high-value sectors and industries. A data analysis project can be separated into four important phases: Project design, where analysts work with stakeholders to identify the business need and translate it into a data project plan; data acquisition and wrangling, where analysts generate, clean, restructure, and otherwise shape data into a useable dataset; data analysis, where analysts use a variety of tools to look for and characterize patterns, and sometimes build predictive models; and data visualization, where they present findings and recommendations as part of a compelling visual narrative that enables effective organizational decision making. Depending on their specific role, analysts may focus on one or more of these phases but should be competent in all.

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Plan a data analysis project to solve a real client problem by identifying project objectives, data requirements, analysis methodologies, success criteria, and key project milestones.	1, 2, 3, 5, 6
2	Create a comprehensive, clean, and well-structured dataset that meets the project requirements.	1, 2, 3, 5, 6
3	Apply data analysis and visualization techniques, as required, to meet project and business objectives.	1, 2, 3, 5, 6
4	Present analysis results, business insights, and recommendations to the client, clearly and professionally.	1, 2, 3, 5, 6
5	Assign roles equitably within the project team to address client need and project requirements.	1, 2, 3, 5, 6
6	Implement strategies for working within a team and maintaining client relationships to facilitate successful project completion.	1, 2, 3, 5, 6

LEARNING PATHWAY

**The time it takes learners to demonstrate competencies will vary. An example of a suggested schedule for learning and development is shown below. Learners will need to plan out their assessment attempts*

within their course. For additional information, please consult the Course Offering Information in Brightspace.

WEEK/HOURS	LEARNING AND DEVELOPMENT PLAN
Week 1	Introduction, group formation
Week 2	Employer presentation, develop team charter
Week 3	Submit team charter, develop project proposal
Week 4	Develop project proposal
Week 5	Submit project proposal
Week 6	Project check-In
Week 7	Develop project solution and presentation
Week 8	Develop project solution and presentation
Week 9	Reading week
Week 10	Project check-In
Week 11	Develop project solution and presentation
Week 12	Develop project solution and presentation
Week 13	Project check-In
Week 14	Develop project solution and presentation
Week 15	Develop project solution and presentation

ASSESSMENT

This course follows an assessment-first approach, in which learners will be assessed, and receive structured feedback, and a personalized learning plan. Learners will also receive differentiated support from an instructor based on their individual needs.

Learners will have a variety of ways to demonstrate they have met the required competency through the demonstration of learning outcomes and criteria as laid out in the rubric. Learners will have multiple (but not unlimited) attempts to prove competency. It is suggested that learners plan out their assessment attempts within their course.

Learners will have flexibility in how they satisfy course learning outcomes while still adhering to the criteria found in the rubric and the Course Offering information. Please refer to the Course Offering Information and the rubric in Brightspace for additional information.

COURSE LEARNING OUTCOMES	ASSESSMENT
1, 2, 3, 4, 5	Work Integrated Learning project

PERFORMANCE STANDARDS

Successful demonstration of all learning outcomes is required to pass this course. This course may be part of a competency or larger program of study.

Please consult with the program area or the Bow Valley College website for completion requirements.

GRADING SCHEME

Symbol	**Description**	**Grade Point Value**
MAST	Mastery	4.00
COMP	Competent	3.00
DEVL	Developing	N/A
F	Fail	0.00
W	Withdraw	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

The capstone project requires the demonstration of tasks from the four competencies from the Data Management and Analytics Post-Diploma Certificate:

- Designing Data Analysis Projects
- Acquiring and Wrangling Data
- Performing Data Analysis
- Visualizing Data and Insights

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or

essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.